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In [ ]: Name : Omkar Hulawale  
Roll_No : 13165  
Batch : A3
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In [11]: import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
In [20]: url = "https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data"  
column_names = ["sepal_length", "sepal_width", "petal_length", "petal_width", "species"]  
df = pd.read_csv(url, header=None, names=column_names)
```

```
In [22]: df
```

```
Out[22]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

```
In [24]: print(df.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
In [32]: df.describe()
```

```
Out[32]:
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [34]: df.describe(include = 'object')
```

```
Out[34]:
```

	species
count	150
unique	3
top	Iris-setosa
freq	50

```
In [36]: df.isnull().sum()
```

```
Out[36]: sepal_length    0
         sepal_width    0
         petal_length    0
         petal_width    0
         species        0
         dtype: int64
```

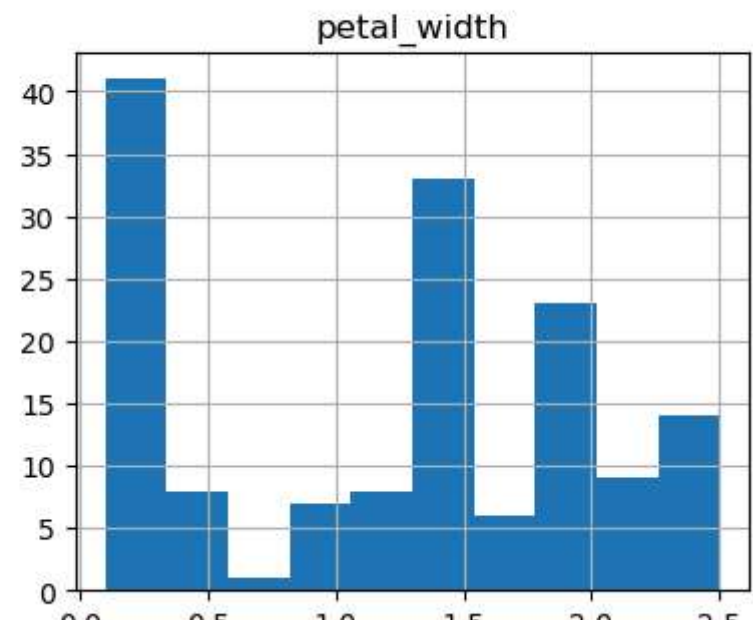
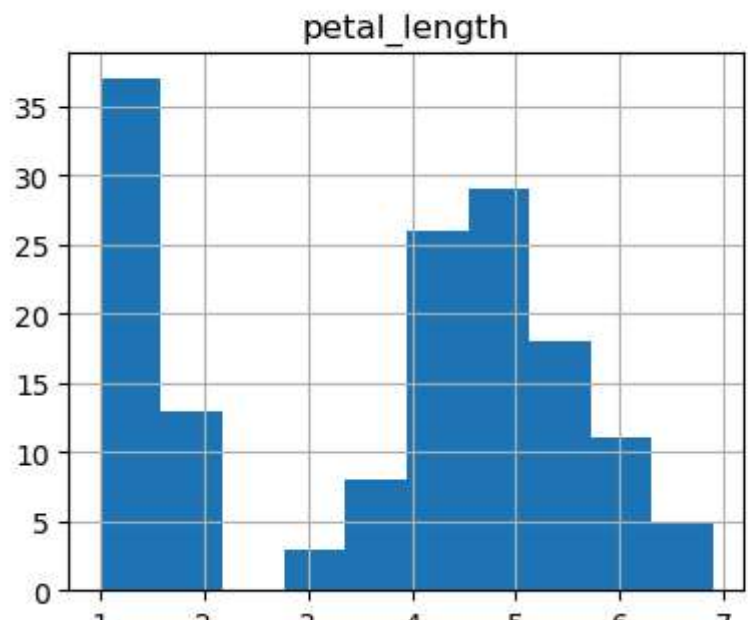
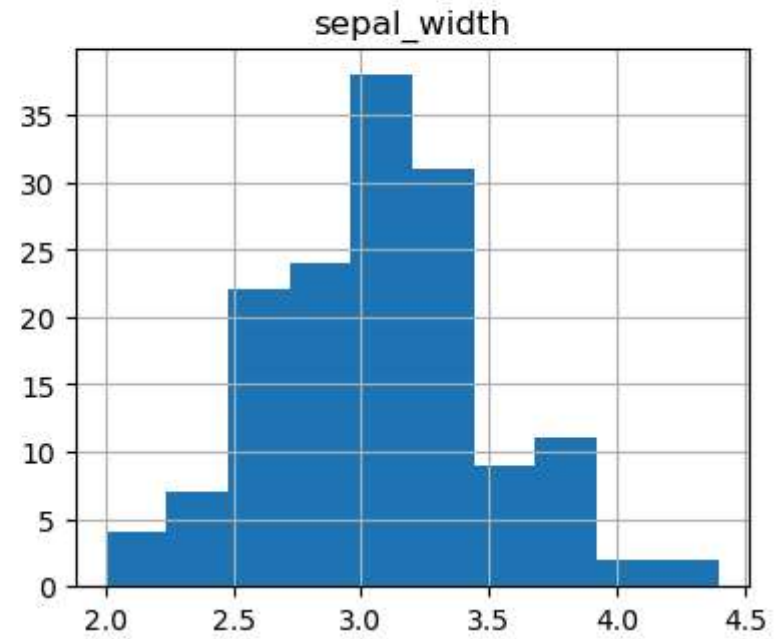
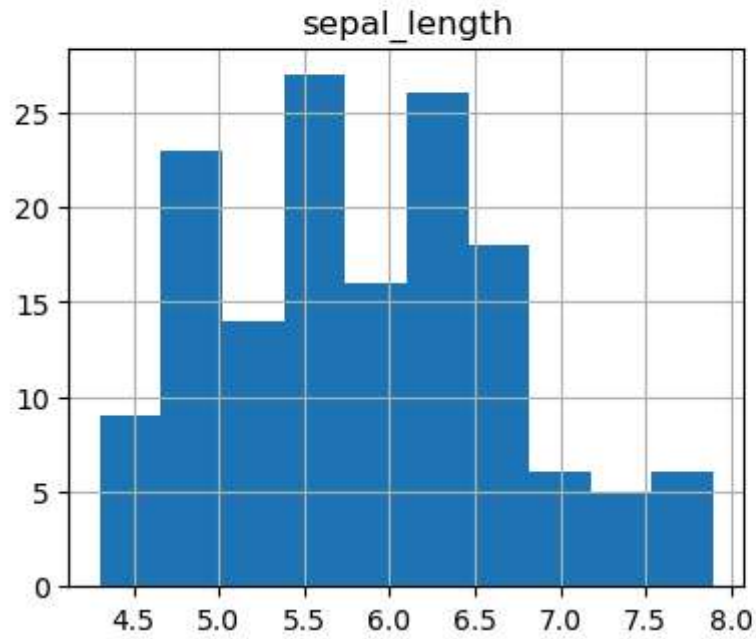
```
In [40]: print("\n\nThe features in the dataset are as follows : ")
         print("1. Sepal length : ", df['sepal_length'].dtype)
         print("2. Sepal width : ", df['sepal_width'].dtype)
         print("3. Petal length : ", df['petal_length'].dtype)
         print("4. Petal width : ", df['petal_width'].dtype)
         print("5. Species : ", df['species'].dtype)
```

The features in the dataset are as follows :

1. Sepal length : float64
2. Sepal width : float64
3. Petal length : float64
4. Petal width : float64
5. Species : object

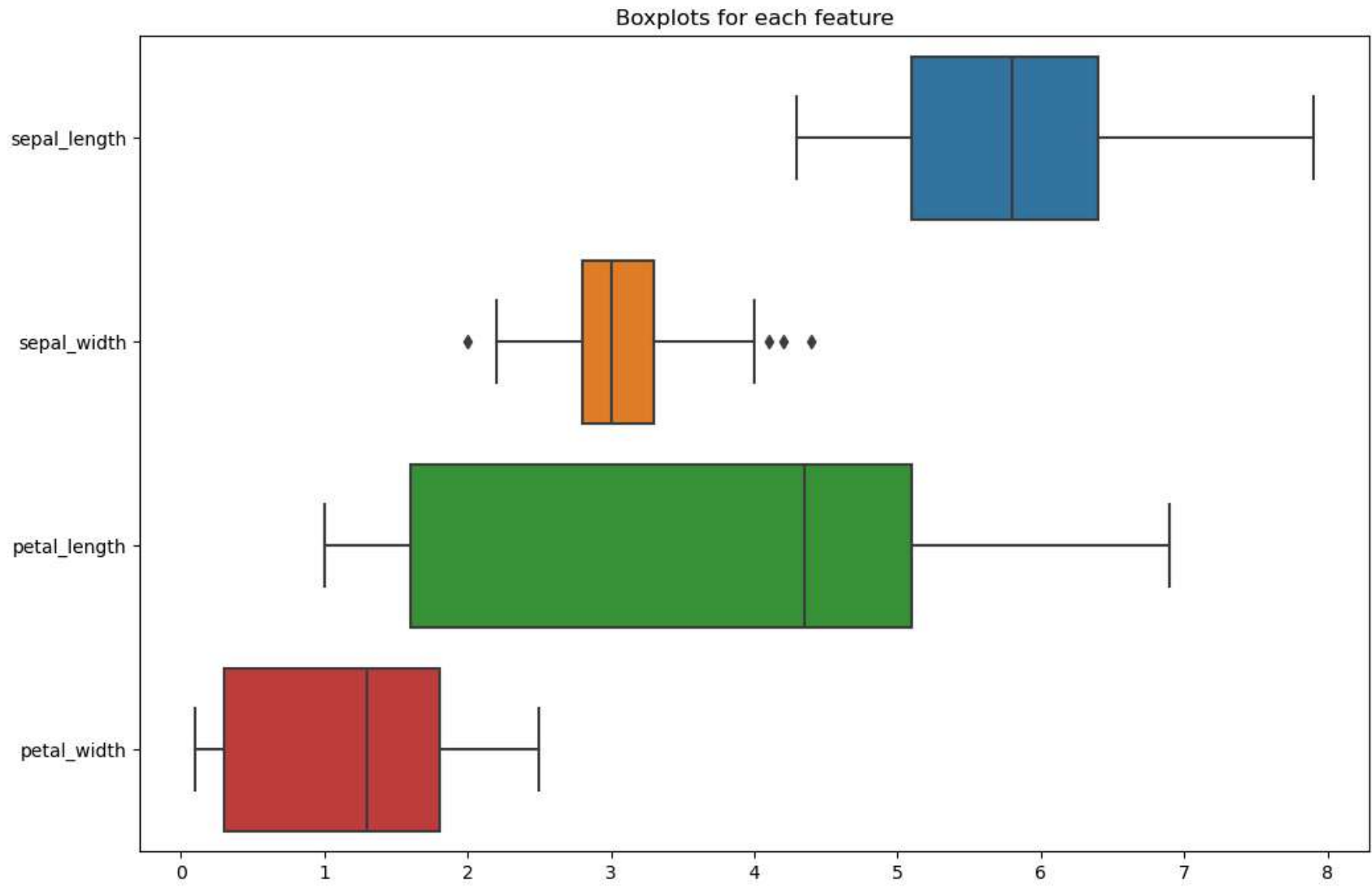
```
In [26]: df.drop('species', axis=1).hist(bins=10, figsize=(10, 8))
         plt.suptitle("Histograms for each feature")
         plt.show()
```

Histograms for each feature



1 2 3 4 5 6 7 0.0 0.5 1.0 1.5 2.0 2.5

```
In [28]: plt.figure(figsize=(12, 8))
sns.boxplot(data=df.drop('species', axis=1), orient='h')
plt.title('Boxplots for each feature')
plt.show()
```



```
In [46]: # Calculate IQR for each feature
Q1 = df.drop('species', axis=1).quantile(0.25)
Q3 = df.drop('species', axis=1).quantile(0.75)
IQR = Q3 - Q1
# Define outliers as values outside 1.5 * IQR
outliers = ((df.drop('species', axis=1) < (Q1 - 1.5 * IQR)) | (df.drop('species', axis=1) > (Q3 + 1.5 * IQR)))
print(outliers.sum()) # This will give the number of outliers in each column
```

```
sepal_length    0
sepal_width     4
petal_length    0
petal_width     0
dtype: int64
```

```
In [68]: # Create a figure with 1 row and 2 columns for the boxplots
fig, axes = plt.subplots(1, 2, figsize=(15, 6)) # 1 row, 2 columns

# Create the first boxplot for 'sepal_length' vs 'species'
sns.boxplot(x='sepal_length', y='species', data=df, ax=axes[0])
axes[0].set_title('Boxplot of Sepal Length by Species')

# Create the second boxplot for 'petal_length' vs 'species'
sns.boxplot(x='petal_length', y='species', data=df, ax=axes[1])
axes[1].set_title('Boxplot of Petal Length by Species')

# Adjust layout for better spacing between the plots
plt.tight_layout()
plt.show()
```

